Team World-33

World-33 High Altitude Balloon Experiment

By: The Montgomery High School Aerospace Club

An Overview

- We will be sending a balloon with various different sensors into the atmosphere
- Data on the composition of gases and the UV radiation will be collected
- After, the relation will be graphed



Project Hypothesis

We predict that the compositions of air with a greater density of the gasses will be more effective in their ability to reduce UV radiation.





Our Inspiration:

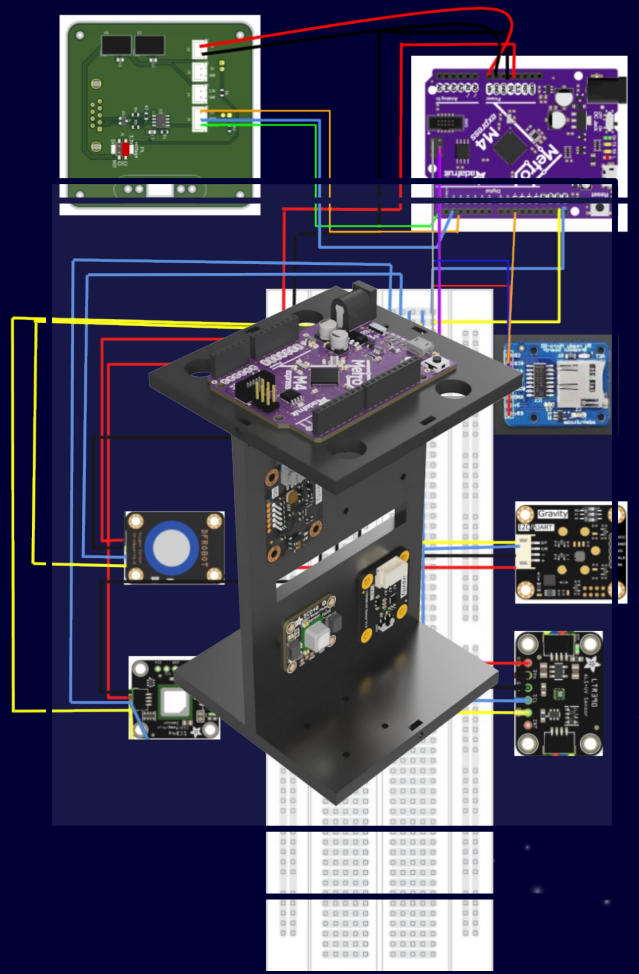


- We wanted to learn more about our atmosphere
- The project seemed feasible for us
- The impact of the project would help many people around the world

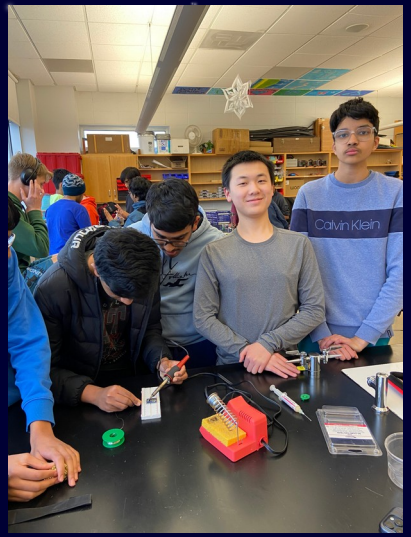
Experiment Design:

We 3D printed a mounting insert for our sensors. We plan to mount the sensors via the holes in the insert.

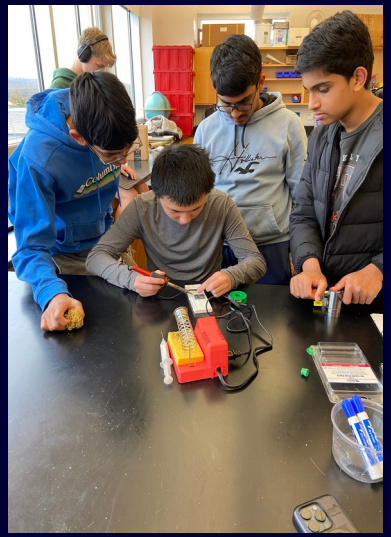
Additionally, we will mount the Metro M4 microcontroller on top. Then, we will solder the wires directly to the sensors and to the microcontroller.

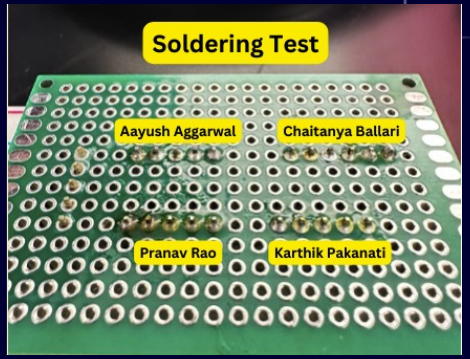


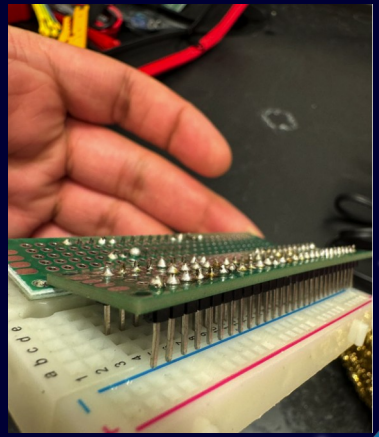
Pictures of our Progress





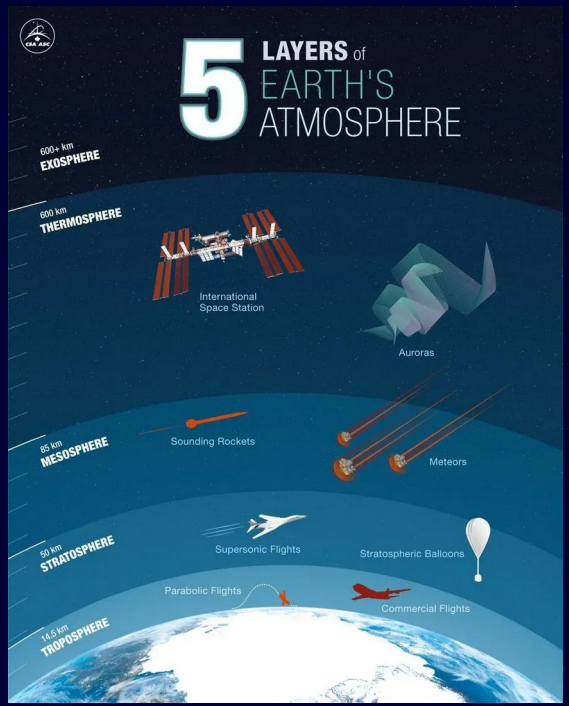






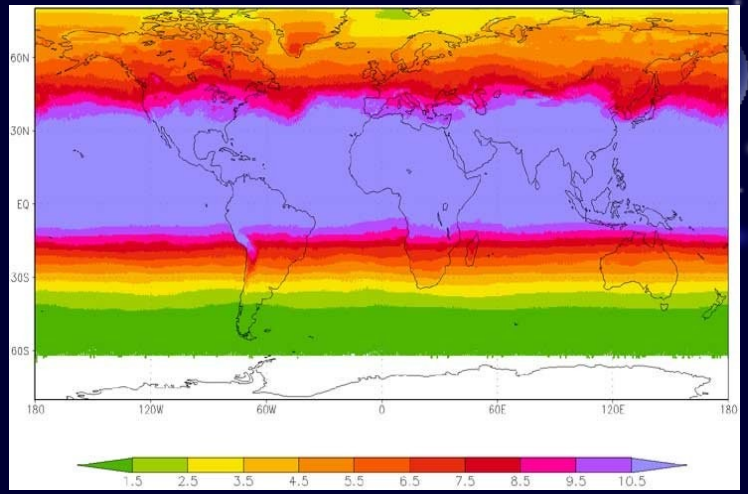
Experiment Specificities

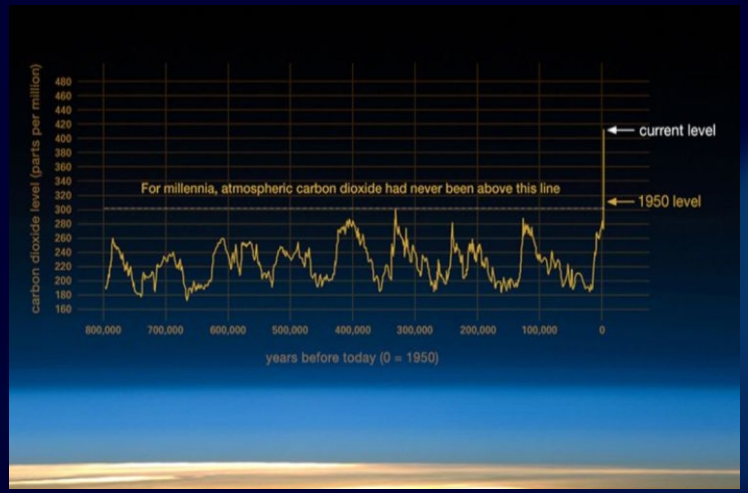
- Our balloon will first go into the atmosphere, sensing gases from an altitude of 1000 ft, all the way up to the stratosphere and ozone layer, about 70,000 ft above
- Using our gas sensors, for NO2, CO2, and Ozone, we will measure the composition of each gas in ppm, and the UV Index at every 1,000 ft
- The sensors will collect data 300 times every 1,000 ft



Impact Of Our Research

- We will be able to find the composition of gasses which blocks UV radiation the most.
- Communities all over the world will try to find solutions to their problems using this gas
- A possible solution they could implement is distributing that gas or composition of gases into the sky to reduce UV radiation.





NASA, Thank You For Listening!

